A Case Study Report

on

Hunting For “Death Stars” In Gaia

From Stars To Planets

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Friday

Abarajitha Devi Krishnamurthy

Student ID: 24023919

Abstract

The stars moving around the space is perfectly normal until it comes too close, straight into the Oort Cloud (e.g. Dybczynski 2002) resulting in serious consequences such as mass extinctions. This case study is about finding stars that could have caused the past extinctions on Earth and a potential death star that might cause one in the future. Further, it includes finding the closest approach of these stars of interest and calculating how much it affected the celestial objects in and around our solar system along with measuring the gravitational unrest caused by the star and how many Long Period Comets (Feng & Bailer-Jones 2015) were launched by it into our solar system. Finally comparing the target stars with the galactic tides to tell which had significant contribution to the LPCs.

Contents List

1.Introduction

Oort cloud is the huge cloud of icy planetesimals surrounding the Sun from about 2000 – 200,000 AU (e.g. Dybczynski 2002). Though no direct evidence is available, its existence is accepted universally. Guessed to be in the form of a sphere, it is thought to be the extreme end of the solar system with many celestial objects of various sizes revolving in it. When a passing star gets too close, the Oort cloud gets disturbed gravitationally leading to the exile of some of the orbiting objects either into the solar system or out. These celestial objects are referred to as Long Period Comets (Feng & Bailer-Jones 2015) which influenced by the gravitational force of the Sun starts to revolve around the sun in a elliptical path. These Long Period Comets can be knocked out of the Oort cloud due to galactical tides too.

Gaia, launched a decade ago, maps the milky way and the celestial objects in it. Monitoring nearly two billion stars, an accurate three-dimensional map of the universe is being created by the Gaia. All of its data are saved in the Gaia archive which we can access and download using ADQL (Astronomical Data Query Language).

In this case study we will search for possible target stars, that caused past mass extinctions and the potential death star that might cause future extinction. After figuring out the formula for the closest approach, we will look into Gaia archive for possible target stars and potential death star, using ADQL, conclude by finding how many Oort cloud objects it influences when compared to the galactic tides that knocks out Long Period Comets regularly.

2. Equations and Techniques

2.1 Relation between Sun and Star

Considering that the star is moving away from the sun, the following diagram is drawn with the below assumptions:

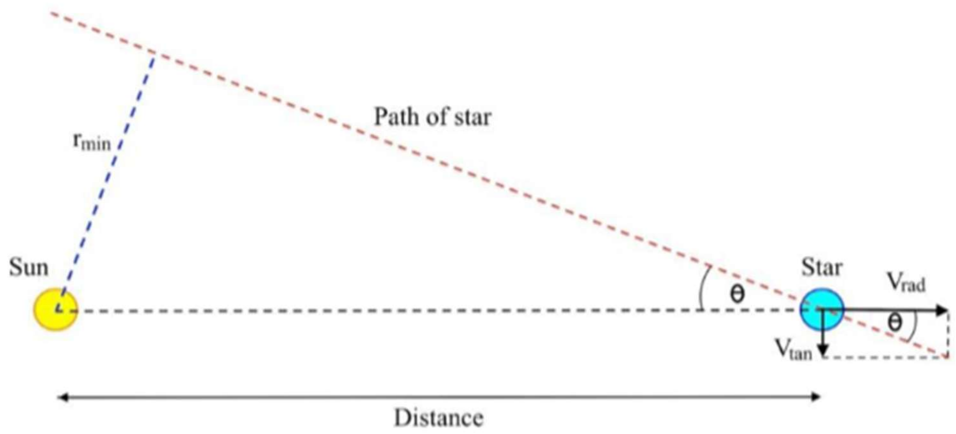
1.The path of star is a straight line

2.The angle between the sun and the path of the star is a small angle θ

3. The radial velocity of the star is V_{rad}

4. The tangential velocity of the star is V_{tan}

5. The minimum distance between the sun and the star at the closest approach, right angle is r_{min}.



(Figure 1) *(Pinfield, Worksheet, 2024)*

The distance between the sun and the path of star is depicted.

Task1:

From the figure, we can see that, tan $\theta = \theta$

$$\theta(rad) = \frac{V_{tan}(km/s)}{V_{rad}(km/s)}$$

where,

rad is radians

Task2:

Converting θ into arcseconds, we have

$$\theta(as) = \frac{V_{tan}(km/s)}{V_{rad}(km/s)} \times 206265$$

[By multiplying 206265 with the value, radians is converted into arcseconds]

Task3:

From the lectures, V_{tan} equation can be obtained

$$V_{tan}(km/s) = 4.74D(pc)\mu_{pm}(as/yr)$$

where,

D is the distance between the sun and the star along the line of sight

μ_{pm} is the proper motion of the star

pc is parseconds

as is arcseconds

On substituting this equation in θ, we have,

$$\theta(as) = \frac{977696.1 \text{ D(pc) } \mu_{pm}(\text{as/yr})}{V_{rad}(km/s)}$$

Task4:

From the lectures, we have

$$r_{min}(AU) = D(pc)\theta(as)$$

Task5:

On substituting θ in r_{min} equation, we have

$$r_{min}(AU) = \frac{977696.1\ D^2(\text{pc})\ \mu_{pm}(\text{as/yr})}{V_{rad}(km/s)}$$

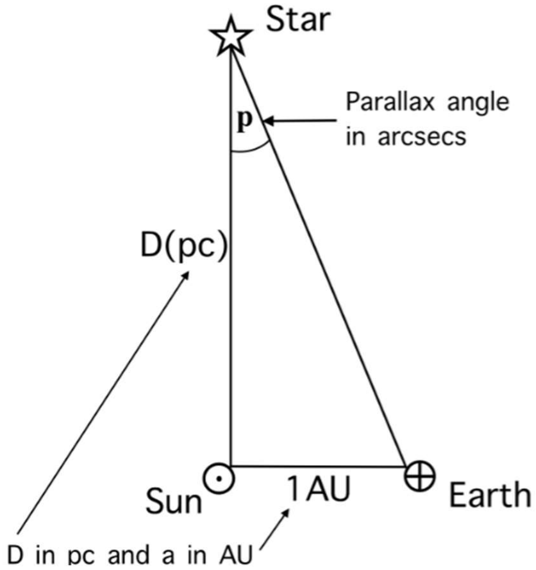


Figure 2 *(Pinfield, Worksheet, 2024)*

The relation between parallax angle and distance shown.

Task6:

By using the above diagram from the lectures, we get

$$D(pc) = \frac{1(AU)}{P(\mathrm{as})}$$

where,

P is the parallax angle

On substituting the parallax equation in r_{min},

$$r_{min}(AU) = \frac{977696.1\ \mu_{pm}(\mathrm{as/yr})}{P^2(as)\ V_{rad}(km/s)}$$

Task7:

The measurements should be converted into Gaia units, so it can be looked up in the Gaia database.

r_{min} in AU,

μ_{pm} in milli arcsecond/year,

P in milli arcsecond/year,

V_{rad} in km/s.

On rewriting the equation with correct units, we have

$$r_{min}(AU) = \frac{977696100\ \mu_{pm}(\text{mas/yr})}{P^2(mas)\ V_{rad}(km/s)}$$

Task8:

Checking the equation with the following example star parameters,

Parallax = 5 mas,

Proper motion = 2 mas/yr,

Radial velocity = 100 km/s,

we get r_{min} as 782156.88 AU. The correct value being 782157 AU, our r_{min} equation is pretty accurate.

Task9:

On converting D from parsec to km, the parallax equation becomes,

$$D(km) = \frac{1000}{P(mas)}\ 3.0857 \times\ 10^{13}$$

Now, by dividing the distance with the star's radial velocity, we arrive at this equation for travel-time of the star,

$$t(seconds) = \frac{D(km)}{V_{rad}(km/s)}$$

where,

t is the star's travel-time in seconds.

On substituting D in the above equation, we have,

$$t_{sec} = \frac{3.0857 \times 10^{16}}{P(mas)\ V_{rad}(km/s)}$$

By dividing it by 31557600, we can convert t from seconds into years,

$$t_{yrs} = \frac{9.777993257 \times 10^8}{P(mas)\, V_{rad}(km/s)}$$

On substituting the example star's parameters, we get

$$t_{yrs} = 1955598.651$$

which can be rounded off to 2 million years, the real travel-time of the example star, proving that our t_{yrs} equation is precise.

2.2 Working with Gaia Database

Task10:

Now we need to find the star that has a r_{min} less than that of Proxima Centauri (1.3 parsec) with the following conditions:

1.Proper motion error of the star less than 0.03 mas/yr

2.Parallax error less than one-tenth of the parallax measurement

3.Radial velocity error is less than one-tenth of the radial velocity measurement.

4.Star's radial velocity is moving away from the sun (positive.)

We can do that by searching the Gaia Archive with the following ADQL query.

ADQL syntax:

SELECT gaia.designation ,

gaia.ra ,

gaia.dec ,

gaia.parallax ,

gaia.parallax_error ,

gaia.pm ,

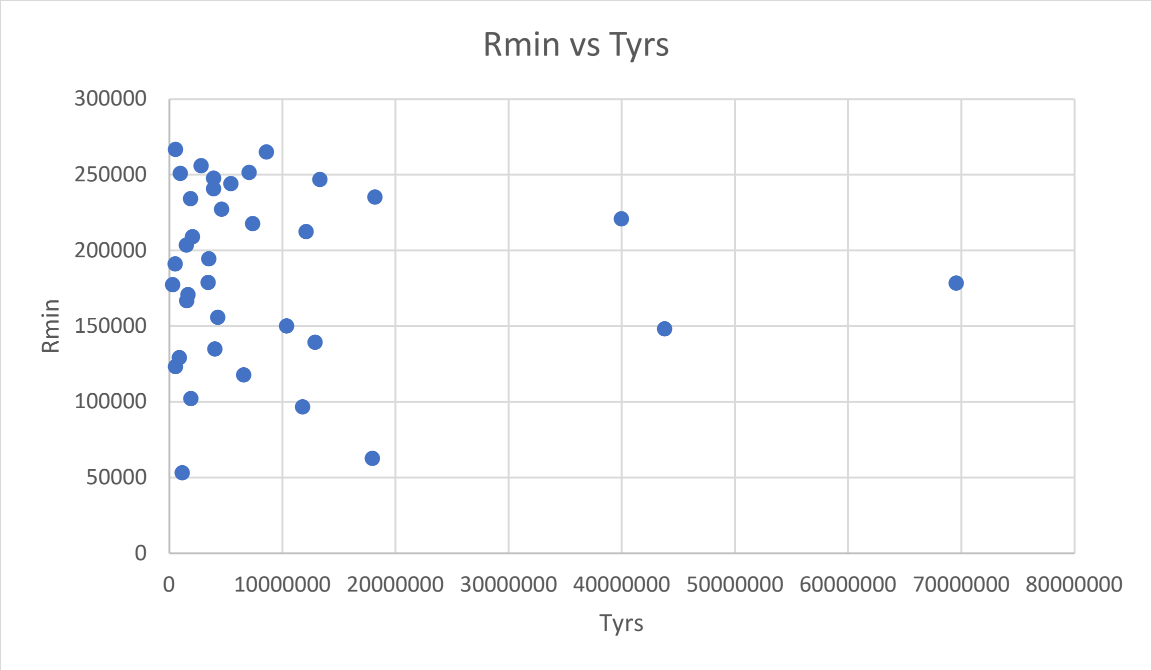
gaia.pmra ,

```
gaia.pmra_error ,
gaia.pmdec ,
gaia.pmdec_error ,
gaia.phot_g_mean_mag ,
gaia.bp_rp ,
gaia.radial_velocity ,
gaia.radial_velocity_error
FROM gaiadr3.gaia_source AS gaia
WHERE gaia.pmra_error < 0.03 AND
gaia.pmdec_error < 0.03 AND
gaia.parallax > 10.0*gaia.parallax_error AND
gaia.radial_velocity > 10.0*gaia.radial_velocity_error AND
gaia.radial_velocity > 0.0 AND
(977696.1*1000.0*gaia.pm)/(gaia.parallax*gaia.parallax*gaia.radial_velocity)
< (1.3*206265.0)
```

On running this query, 36 stars will be detected by the Gaia Archive.

Task11:

The CSV file downloaded has 36 stars with the following r_{min} and t_{yrs} information in it.



Graph 1 *– The 36 stars that are moving away from the Sun are plotted in the* r_{min} *-* t_{yrs} *graph*

2.3 Searching for Target Stars

Task12:

Using Kepler's law, we have

$$P^2(years)M(solar\ mass) = a^3(AU)$$

where,

P is the time period taken by Oort cloud to revolve around the sun in an elliptical orbit

M is the mass of the sun

a is the semi major axis of the elliptical path

We know that the aphelion and the perihelion of the Oort cloud object is 80000 AU and 1 AU respectively and therefore, the semi major axis can be calculated by halving the sum of perihelion and aphelion.

$$a(AU) = \frac{80000 + 1}{2}$$

$$a(AU) = 40000.5$$

The mass of the main star that is our sun is 1 AU, substituting this information on Kepler's equation,

$$P^2(yrs) \times 1 = 40000.5^3$$

$$P^2(yrs) = 64.0024 \times 10^{12}$$

$$P(yrs) = \sqrt{64.0024 \times 10^{12}}$$

$$P(yrs) = 8.0001 \times 10^6$$

The time taken for the Oort cloud to travel from aphelion to perihelion is,

$$\frac{P}{2}(yrs) = 4.000075 \times 10^6$$

The lagging time is approximately 4 million years.

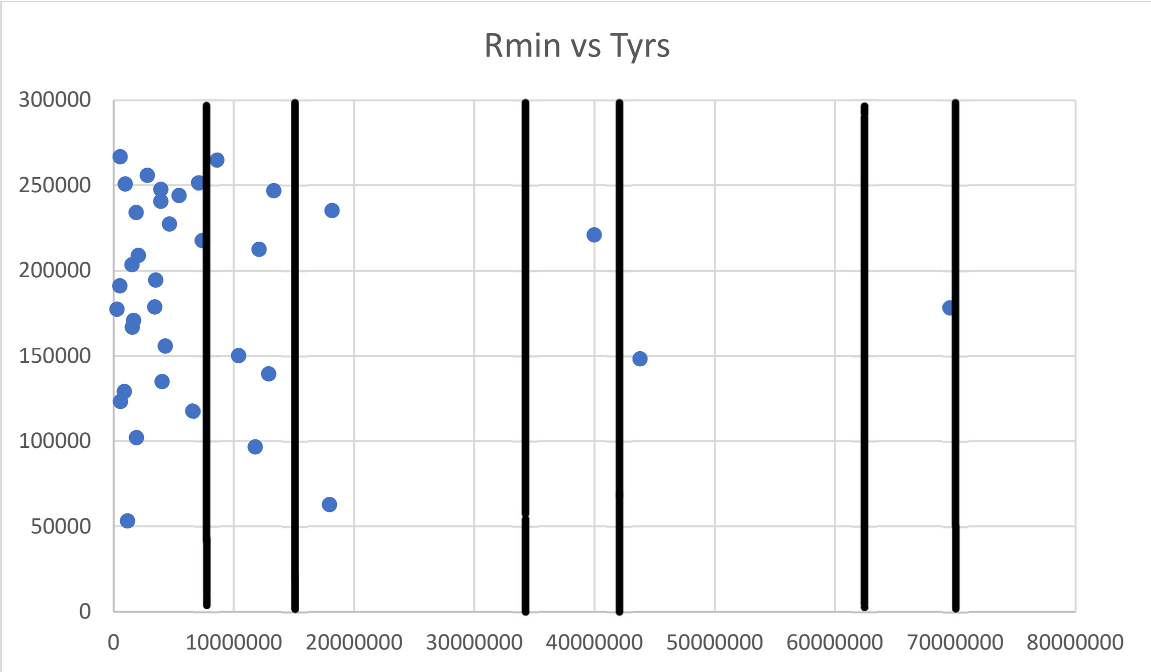
Task13:

Extinction	Age (millions of years)
Middle Miocene	11.6
Mid-late Eocene	37.8
End Cretacious	66
Cenomanian/Turonian	93.9

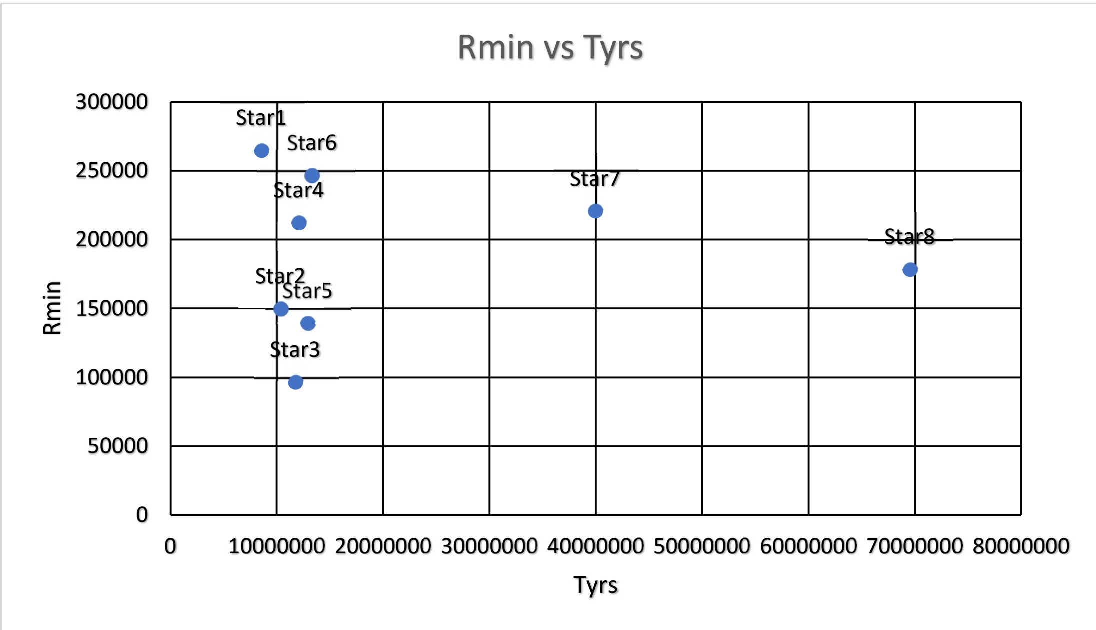
Table 1. *The mass extinctions of the past and the timeline*

(from Rampino & Caldeira 2015)

Considering the above table of mass extinctions, along with an error margin of 4 million years, we can narrow down our targets to just 8 stars, as marked in the graph below.



Graph 2 *– Black lines tells the timeline of the extinction (time lag included) and the points inside depict the stars that passed by during that timeline.*



Graph 3 *– Target stars plotted in a separate graph*

Stars	Gaia designation	Right ascension (ra)	Declination (dec)
Star 1	Gaia DR3 6676301403484000896	309.9039769	-42.82336028

Star 2	Gaia DR3 3113059336290899328	105.3344897	0.692714488
Star 3	Gaia DR3 2553954478106385664	7.72270889539023	3.89840701281009
Star 4	Gaia DR3 5607782706396947328	105.900095723058	-29.9182409212418
Star 5	Gaia DR3 5083233196571443968	59.3590891945519	-24.7416420266379
Star 6	Gaia DR3 5529728406003908864	59.3590891945519	-38.2970894327986
Star 7	Gaia DR3 3048814219048540672	109.520199748279	-8.18813864521128
Star 8	Gaia DR3 4052494195752756992	273.256342375173	-27.1915303077413

Table 2 *– Information about target stars – Gaia designation, ra, dec*

Stars	r_{min}	t_{yrs}	Possible Mass Extinction
Star 1	264977.8358	8582597.779	Middle Miocene
Star 2	150197.8271	10378636.55	Middle Miocene
Star 3	96654.09281	11773771.14	Middle Miocene
Star 4	212428.5051	12107064.89	Middle Miocene
Star 5	139448.595	12901414.32	Middle Miocene
Star 6	246884.7298	13311413.84	Middle Miocene
Star 7	220885.1398	39972683.04	Mid-late Eocene
Star 8	178472.627	699558560.74	End Cretaceous

Table 3 *– Information about target stars -* r_{min}, t_{yrs} *and possible mass extinction*

The above tables (Table 2 & 3) tell us about the Gaia designation of the star and its right ascension and declination along with r_{min}, t_{yrs} and the mass extinctions they could have caused.

2.4 Seeking Death Star

Task14:

Potential death star can be identified by reversing the direction of the star's radial velocity (negative) making the Gaia archive find stars that are moving towards the sun.

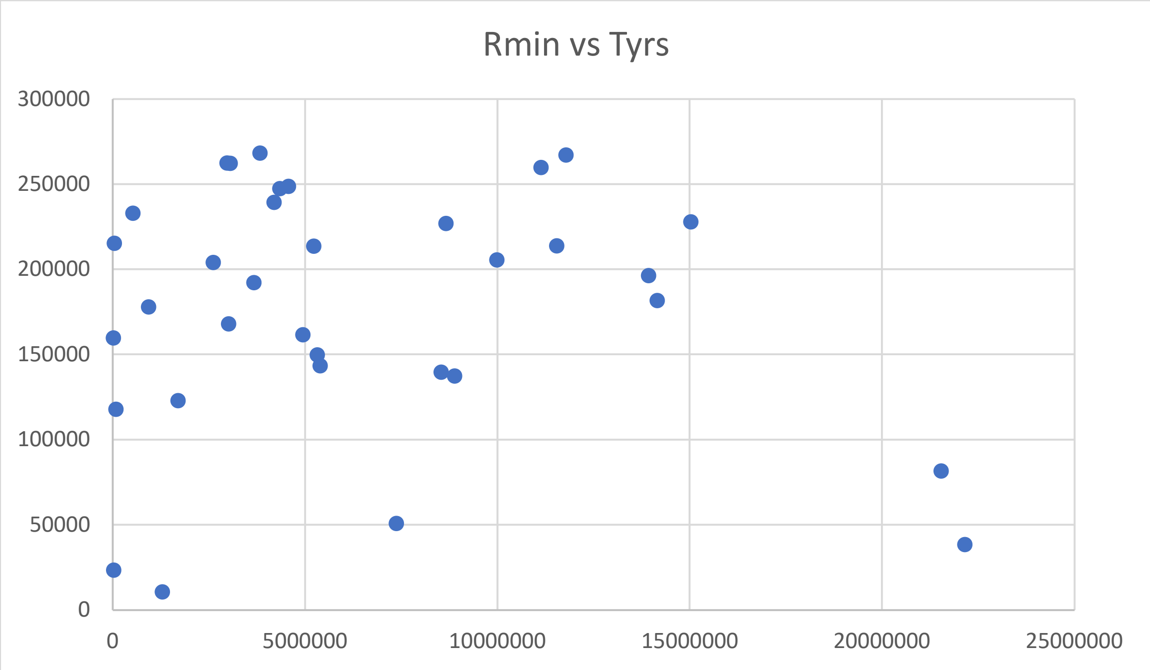
ADQL Syntax:

```
SELECT gaia.designation ,
gaia.ra ,
gaia.dec ,
gaia.parallax ,
gaia.parallax_error ,
gaia.pm ,
gaia.pmra ,
gaia.pmra_error ,
gaia.pmdec ,
gaia.pmdec_error ,
gaia.phot_g_mean_mag ,
gaia.bp_rp ,
gaia.radial_velocity ,
gaia.radial_velocity_error
FROM gaiadr3.gaia_source AS gaia
WHERE gaia.pmra_error < 0.03 AND
gaia.pmdec_error < 0.03 AND
gaia.parallax > 10.0*gaia.parallax_error AND
(-1)*gaia.radial_velocity > 10.0*gaia.radial_velocity_error AND
gaia.radial_velocity < 0.0 AND
(977696.1*1000.0*gaia.pm)/(gaia.parallax*gaia.parallax*gaia.radial_velocity*(
-1.0)) < (1.3*206265.0)
```

This query detected 34 stars.

Task15:

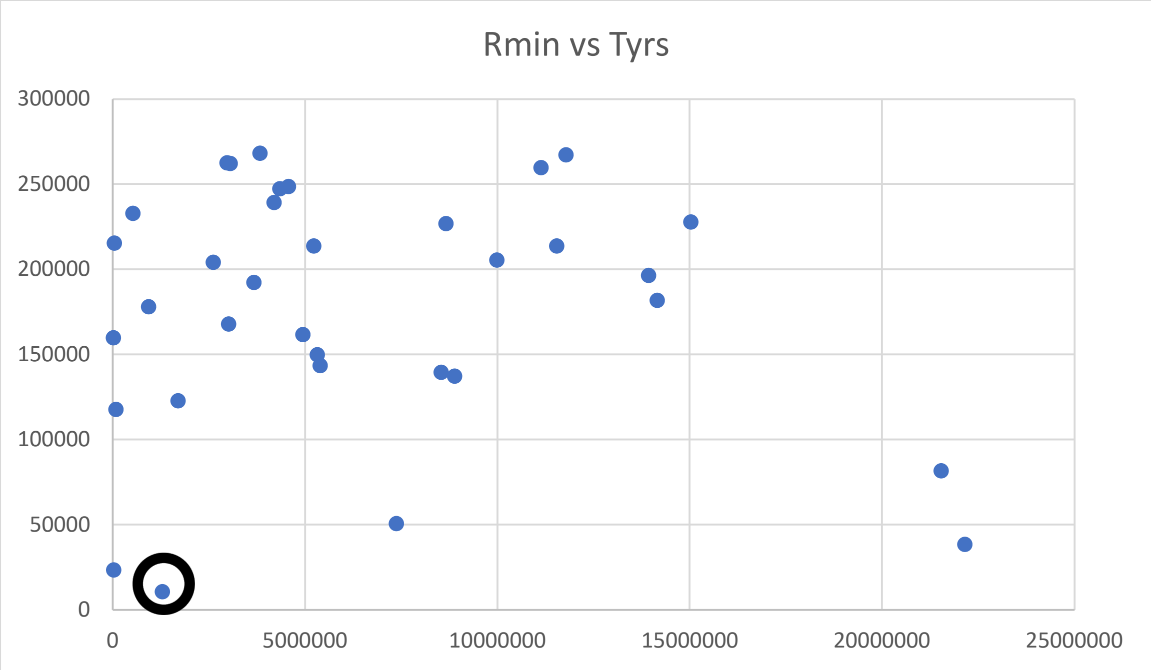
The CSV file downloaded has stars with the following r_{min} and t_{yrs} :



Graph 4 *– Scatter plot about 34 stars moving towards the Sun*

Task16:

The potential death star is a star with least r_{min}.



***Graph 5** – The circled star has the least r_{min}*

Let this star be Star 9.

Star 9 will reach a distance of 10571.07 AU from the sun and it might take 1.2 million years to reach such a close distance.

Our potential death star, Star 9 has GJ number 710.

Task17:

Fractional error on parallax, proper motion and radial velocity can be calculated using the formula, $2\frac{\delta P}{P}$, $\frac{\delta \mu_{pm}}{\mu_{pm}}$ and $\frac{\delta V_{rad}}{V_{rad}}$ respectively, where

$\delta \mu_{pm}$ is the error in proper motion,

δV_{rad} is the error in radial velocity,

δP is the parallax error.

The data regarding the death star can be used to calculate its fractional errors.

Parameters of potential death star	**Value**
Parallax	52.3962922052905
Parallax error	0.0171099

Proper motion	0.42803693
Proper motion error ra Proper motion error dec	0.019432956 0.016565036
Radial velocity	-14.419985
Radial velocity error	0.2563099
r_{min}	10571.07621

***Table 4** – Parameters of the death star*

Their values are as follows:

$$2\frac{\delta P}{P} = 0.000653096$$

$$\frac{\delta \mu_{pm}}{\mu_{pm}} = 0.042050101$$

$$\frac{\delta V_{rad}}{V_{rad}} = -0.01777463$$

Task18:

Fractional errors for the death star can be found using the following formula

$$\frac{\delta r_{min}}{r_{min}} = \sqrt{\left(\frac{\delta \mu_{pm}}{\mu_{pm}}\right)^2 + \left(\frac{\delta V_{rad}}{V_{rad}}\right)^2 + \left(2\frac{\delta P}{P}\right)^2}$$

where,

δr_{min} is the error in r_{min}

Using the above data of the death star, we get the fractional error on r_{min}

$$\frac{\delta r_{min}}{r_{min}} = 0.045657146$$

Task19:

The actual error of r_{min} was calculated to be,

$$\delta r_{min} = 482.6451695$$

$$\sigma = 482.6451695$$

where,

σ is the standard deviation

The 99.74% confidence limit is,

$$3\sigma = 1447.9355085$$

Task 20:

Now, r_{min} ranges between the following upper and lower limits for 3σ .

$$upper\ limit = 12019.0117185$$

$$lower\ limit = 9123.1407015$$

Task21:

Using a standard-normal-distribution table, we can find that the confidence level of 3 times the error bar is 99.74%, meaning the percentage of the value being lesser than lower limit is 0.13%

The probability that the closest approaching star is below the lower limit is 0.0013, states that our prediction is accurate.

2.5 Stellar Mass Estimation

Task22:

From the lectures, we know that the distance modulus equation is,

$$m_g + M_g = 5logD - 5$$

where,

m_g is the apparent magnitude

M_g is the absolute magnitude

D is the distance between two stars

On substituting m_g as G and D as $\frac{1000}{P}$ and rewriting, we have

$$M_g = G - 10 + 5logP$$

where,

G is the apparent magnitude

P is the parallax angle

On substituting the values of stars, we get the following M_g values

Stars	M_g
Star 1	3.092572461
Star 2	4.426855763
Star 3	2.602981955
Star 4	4.889090351
Star 5	5.367367152
Star 6	4.444115676
Star 7	1.113625343
Star 8	0.125009926
Star 9	7.659590777

Table 5 *– Absolute magnitude of the target stars*

Task23:

With the available information and comparing it with the Gaia colour absolute magnitude diagram we can see the star types

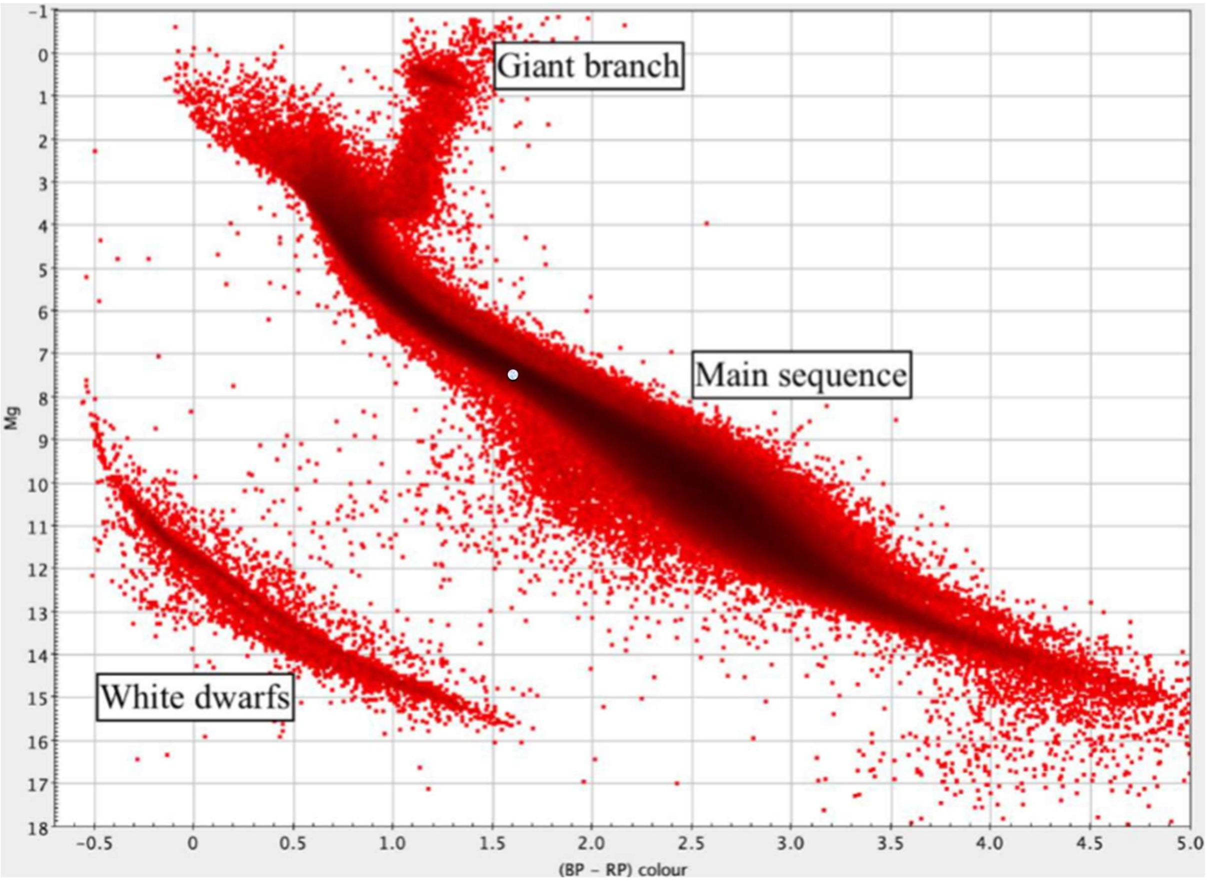


***Figure 3** – Gaia colour absolute magnitude diagram*

Stars	Star type
Star 1	Main Sequence
Star 2	Main Sequence
Star 3	Main Sequence
Star 4	Main Sequence
Star 5	Main Sequence
Star 6	Main Sequence
Star 7	Giant Branch
Star 8	Giant Branch
Star 9	Main Sequence

***Table 6** – Star type of each target star*

Task24:

White dwarfs have a mass range of 0.4 to 0.7 M_{sun} and giant stars have a mass range of 0.3 to 8 M_{sun}. The mass range of the main sequence stars can be found using linear interpolation from the table of established M_g and mass values for main-sequence stars.(Appendix)(Pinfield, Worksheet, 2024)

Using the linear interpolation formula, (Pinfield, Worksheet, 2024) we can find the mass of the interesting stars.

$$y = y_0 + (x - x_0)\frac{y_1 - y_0}{x_1 - x_0}$$

Considering that x is the absolute magnitude of the interesting stars, and y being the mass of the stars, we can conclude the following.

Stars	$M_g(x)$	x_0	x_1	y_0	y_1
Star 1	3.09	2.99	3.10	1.44	1.38
Star 2	4.4268	4.426	4.635	1.03	1.00
Star 3	2.60	2.51	2.69	1.61	1.50
Star 4	4.889	4.801	4.914	0.98	0.97
Star 5	5.367	5.34	5.553	0.90	0.88
Star 6	4.444	4.426	4.635	1.03	1.00
Star 9	7.65	7.57	7.74	0.64	0.62

***Table** 7 – Values given for the linear interpolation*

Stars	Mass (solar masses)
Star 1	1.384051385
Star 2	1.029877163
Star 3	1.553177694
Star 4	0.972204394
Star 5	0.897430314
Star 6	1.027399664
Star 7	0.3 – 8
Star 8	0.3 – 8
Star 9	0.629459909

***Table 8** – Stellar masses of target star in Solar masses*

Task25:

Now using the following equation, we can find the r_p ;

$$r_p = \sqrt{a^2 + b^2} + a$$

where,

a is the semi major axis of the hyperbolic path b is the semi minor axis of the hyperbolic path

r_p is the actual closest distance the star reaches when it travels in a hyperbolic path

b is the r_{min} and a can be calculated by the formula ;

$$a = \frac{G(\ M_{star} + M_{sun})}{V_{star}^2}$$

where,

G is the gravitational constant $6.67 \times 10^{-1}\ \ Nm^2Kg^{-2}$

V_{star} is the velocity of the star given by the formula

$$V_{star}^2 = V_{rad}^2 + V_{tan}^2$$

$$V_{tan}(km/s) = 4.74D(pc)\mu_{pm}(as/yr)$$

On substituting $D(pc) = \frac{1(AU)}{P(\text{as})}$ we have,

$$V_{tan}(km/s) = 4.74\frac{1}{P(as)}\mu_{pm}(as/yr)$$

Using the V_{tan} formula and the star's information we can get the V_{star} value. Using that and other information such as a and b we get the following;

Stars	$V_{star}(km/s)$	a	r_p
Star 1	14.31059843	-1.5444×10^{12}	3.9612×10^{16}
Star 2	72.89747228	-5.0676×10^{10}	2.2454×10^{16}
Star 3	19.3439414	-9.0498×10^{11}	1.4448×10^{16}
Star 4	49.69393661	-1.0595×10^{11}	3.1758×10^{16}
Star 5	31.89763915	-2.4740×10^{11}	2.0847×10^{16}
Star 6	52.53941357	-9.7438×10^{10}	3.6909×10^{16}
Star 7	52.8082795	-6.1844×10^{10} to -4.2815×10^{11}	3.3022×10^{16} to 3.3021×10^{16}

Star 8	35.60749208	-1.3602×10^{11} to -9.4171×10^{11}	2.6681×10^{16} to 2.6680×10^{16}
Star 9	14.42003699	-1.0396×10^{12}	1.5793×10^{15}

***Table 9** – Value of* r_p

2.6 Influence on Oort Cloud Objects

The gravitational agitations caused by the approaching star can be calculated using the formula

$$g_{proxy} = \frac{M_{star}}{r_p^2 \times V_{star}}$$

where,

g_{proxy} is the proxy measure used to enumerate the strength of the gravitational agitations.

Task26:

The g_{proxy} in the given equation is similar to the gravitational force equation, directly proportional to the mass (M_{star}) and inversely proportional to the square of the distance (r_p^2).

$$F = G\frac{Mm}{r^2}$$

Task27:

The g_{proxy} values/ranges for the interesting stars are as follows

Stars	g_{proxy}
Star 1	0.058685958
Star 2	0.02667919
Star 3	0.366150837
Star 4	0.018469559
Star 5	0.061638098
Star 6	0.013667625
Star 7	0.004960356 to 0.132279105
Star 8	0.011268499 to 0.300511461

Star 9	16.66324917

Table 10 - *g_{proxy} values for target stars*

Task28:

The number of Oort cloud objects thrown into the solar system as a result of the gravitational unrest is denoted as N_e. It has the following relation with the g_{proxy} (Pinfeild, Worksheet, 2024)

$$N_e = 182g_{proxy}$$

The values of N_e for the interesting stars are given in the below table;

Stars	N_e
Star 1	10.6808444
Star 2	4.855612522
Star 3	66.63945228
Star 4	3.361459693
Star 5	11.21813392
Star 6	2.487507719
Star 7	0.902784862 to 24.07479708
Star 8	2.050866863 to 54.69308598
Star 9	3032.711349

Table 10 - *N_e values for target stars*

Task29:

A number of long period comets set off due to other reasons besides a star intruding the Oort cloud, one major reason being the galactic tides with a record of 1900 every million years. On comparison, our potential death star has triple the amount of LPCs, 3032.

Task30:

Straight line as the path of the star (r_{min}) was the main assumption in this case study, but it didn’t have much impact on the results as it was corrected halfway into a hyperbolic path (r_p) and also because the difference between the two was negligible. The error of the stars was not taken into account in this case study because the error margin itself was small and the difference because of it will not affect the results adversely.

3. Conclusion

Starting with the assumption that the star approaches sun in a straight line, equation for the closest approach of the star is written. In addition, a ADQL syntax, meeting all the necessary conditions is used to obtain the data of 36 stars, from the Gaia archive, moving away from the sun, significantly narrowing the scope to find the target stars. On comparison with the past extinction timeline, keeping in mind the lag of 4 million years, 8 suitable stars were identified. The potential death star was identified by reversing the radial velocity direction then choosing a star with the least r_{min} value among the second Gaia archive data of 34 stars, with the GJ number 710. Finally, the actual closest distance, when the path is hyperbolic, is calculated, after which the gravitational unrest and its influence on the Oort Cloud, number of LPCs launched, is measured.

4. References

Dybczynski, P. A., 2002, A&A, 396, 283.

ESA - Gaia overview

Feng, F., Bailer-Jones, A. L., 2015, MNRAS, 454, 3267

Pinfield, Worksheet, 2024

Rampino, M. R., Caldeira, K., 2015, MNRAS, 454, 3480

Vallenari, A., et al., 2022, A&A, AA43940-22

5. Appendix

Masses and Gaia absolute magnitudes for main sequence stars

M/M_{sun}	M_g (Gaia absolute G mag)
3.38	-0.01
2.75	0.515
2.68	0.615
2.18	1.00
2.05	1.16
1.98	1.345
1.86	1.69
1.93	1.92
1.88	1.98
1.83	2.09
1.77	2.19
1.81	2.27
1.75	2.37
1.61	2.51
1.50	2.69
1.46	2.89
1.44	2.99
1.38	3.10
1.33	3.26
1.25	3.56
1.21	3.66
1.18	3.90
1.13	4.105
1.08	4.195
1.06	4.325
1.03	4.426
1.00	4.635
0.99	4.703
0.985	4.757
0.98	4.801
0.97	4.914
0.95	5.006
0.94	5.098

0.90	5.34
0.88	5.553
0.86	5.65
0.82	5.83
0.78	6.20
0.73	6.53
0.70	6.83
0.69	7.02
0.64	7.57
0.62	7.74
0.59	8.03
0.57	8.16
0.54	8.44
0.50	8.82
0.47	8.98
0.44	9.29
0.40	9.67
0.37	10.05
0.27	10.87
0.23	11.21